



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

March / April 2021

INFORMATION COMMUNICATION AND TECHNOLOGY (N540)

TIME: 1 HOUR 45 MINUTES

Candidates answer on the Question Paper.

INSTRUCTIONS TO CANDIDATES

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Any rough working should be done in this booklet.
5. Do NOT use calculators.
6. Answer **ALL** questions.
7. This document consists of **9** questions printed on **20** pages
8. The number of marks is given in brackets [] at the end of each question or part question.
9. The total of the marks for this paper is **100**.

For Examiners' use											
Question No.	1	2	3	4	5	6	7	8	9	Total	Signature
Marker											
Team Leader											
Quality Controller											
CE/ACE											

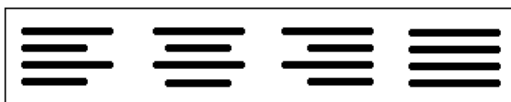
Section A

Question 1 (15 Marks)

Circle the letter corresponding to the correct answer.

Each item carries one mark.

- (a) In a word processor, what is the function of the following icons?



- A Set the line spacing
- B Add extra lines
- C Copy and paste the text
- D Change alignment

- (b) Which of the following is a search engine?



- (c) In a spreadsheet, which of the following represents the three parts that make up an **IF** function?

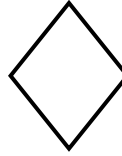
- A CONDITION, TRUE, FALSE
- B CONDITION, RANGE, CRITERIA
- C CONDITION, OPERATORS, CRITERIA
- D CONDITION, FUNCTION, RANGE

- (d) In a flowchart, which of the symbols below is used to insert the message 'Display Sum'?

A



B



C



D



- (e) *Disk Defragmenter* is a utility software designed to _____ .

- A reboot the computer system
- B share files
- C rearrange stored files
- D decrease access speed

- (f) A _____ is a memory location that can be given a symbolic name.

- A counter
- B sequence
- C variable
- D construct

- (g) **One** of the tasks of memory management is to _____ .

- A allow the user to interact with the computer system
- B deal with loading and running of application programs
- C allocate a time slice to each job to be processed
- D manage the transfer of data and files

(h) The Data Protection Act is a law designed to protect _____.

- A** personal information
- B** computers from hackers
- C** copyright materials
- D** ownership rights

(i) An antivirus software _____.

- A** detects and removes malware
- B** retrieves deleted files
- C** verifies user names and passwords
- D** optimises drives

(j) Which one of the following uses radio waves to provide wireless high-speed internet?

- A** HTML
- B** ISP
- C** WIFI
- D** RAM

(k) The word processing feature which outlines the chapters or section names with their corresponding page numbers in a document is the _____.

- A** Styles tab
- B** Page Break
- C** Font style
- D** Table of Contents

- (l) Which feature in a spreadsheet is used to replicate a cell's content?
- A** Conditional formatting
 - B** Fill handle
 - C** Cell address
 - D** Working style
- (m) What is the format of 'th' in the date '**20th March**'?
- A** Italics
 - B** Underline
 - C** Subscript
 - D** Superscript
- (n) A MODEM is _____.
- A** a software that prevents unauthorised access to a private network
 - B** a hardware that allows a computer to send and receive data
 - C** an online service allowing users to stay in touch
 - D** a company that provides access to the internet
- (o) The physical arrangement of a network is called a network _____.
- A** interface
 - B** drive
 - C** topology
 - D** security

Question 2 (10 Marks)

Fill in the blanks with the correct word from the list given below.

report	podcasting	troubleshooting	hardware	ownership	HTML
paraphrasing	router	filter	dialogue	template	decision

- (a) A device that allows the user to access the internet from two or more devices is a _____.
- (b) The transfer of data from input devices to computer memory is performed by _____ management.
- (c) Identifying and resolving a problem in a computer system is called _____.
- (d) A sample document that has been pre-formatted is a _____.
- (e) In a spreadsheet, a _____ is used to extract the data so that only the information needed is displayed.
- (f) Keeping the same meaning of an original text without copying the same words is called _____.
- (g) A computer language used to create websites is _____.
- (h) The legal right for the creation of resources on the internet is known as _____.
- (i) A _____ symbol is used to show a condition which may have two possible outcomes in a flowchart.
- (j) A free web service that makes audio files available for streaming and downloading is known as _____.

Question 3 (10 Marks)

Tick (✓) **True** or **False** next to each of the statements below.

An **example** has been provided.

		True	False
(a)	Disk formatting is an example of a utility program.	✓	
(b)	A paper jam prevents a printer from working properly.		
(c)	Vodcasting is a web service for streaming and downloading video files.		
(d)	In processor management, a time slice is given to each job that needs to be processed.		
(e)	Radiation emitted by the screen may cause Repetitive Strain Injury (RSI).		
(f)	In a COUNTIF function 'Criteria' refers to the conditions that must be met for the cells to be counted.		
(g)	People do not need to sign in to join a chat room.		
(h)	A sequence is used to refer to steps that are repeated many times.		
(i)	<i>WordArt</i> can be used to add an artistic flair to your text.		
(j)	<i>DupliChecker</i> is an antivirus software.		
(k)	Styles are applied to a range of cells to keep the formatting of the worksheet consistent.		

Question 4 (10 Marks)

- (a) Match **Column A** to **Column B** by writing down the corresponding number in the **Answer Grid** below. An **example** has been provided in the Answer Grid.

Column A		Column B	
(i)	Single-user system	1	is the science of how we interact with the objects around us.
(ii)	Multi-user system	2	is used to recover data in case of loss.
(iii)	Ergonomics	3	supports only one user at a time.
(iv)	Multi-processing system	4	refers to a powerful computer that manages communication around a network.
(v)	Server	5	is a single page of a presentation.
(vi)	References Tab	6	allows several users to access a computer's resources at the same time.
(vii)	Data backup	7	provides us with the option of creating a table of content in MS word
		8	is capable of supporting and utilising more than one computer processor.

Answer Grid

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
3						

[6]

- (b) Write down the most appropriate type of network for each of the given characteristics.

	Characteristic	Most appropriate type of network
(i)	Designed for a town or city	
(ii)	Connecting computers across a country or around the World	
(iii)	Limited to a single site	
(iv)	Limited to a small office or room	

[4]

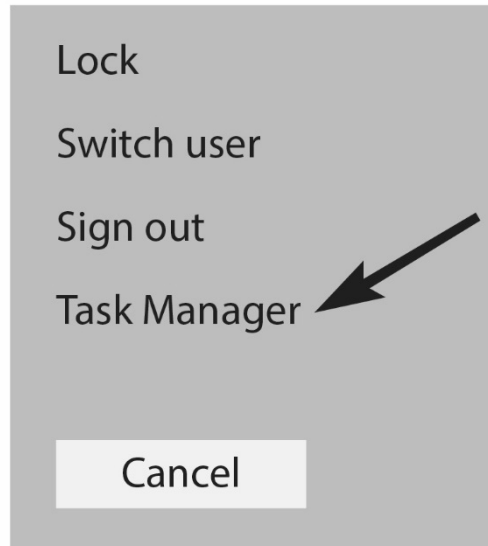
Question 5 (8 Marks)

- (a) **Tick (✓)** the **three** (3) possible solutions for the problem of 'Accidental deletion of a file'.

	Solution	Tick (✓)
1	Check that all cables are properly connected.	
2	Open the Recycle Bin. Right-click on the file and click on Restore.	
3	Change the batteries of the wireless mouse or keyboard.	
4	Press Ctrl + Z on the keyboard to retrieve the file.	
5	Right-click on the icon where file was located. In the pop-up menu, select "Undo Delete".	
6	Turn the computer off and on again.	

[3]

- (b) Which **three keys** should be pressed at the same time on the keyboard to open the menu shown below?



++

[3]

- (c) (i) Give **one** possible cause of a blank screen when a computer is still on.

.....

.....

[1]

- (ii) Suggest **one** solution to the cause mentioned in (c) (i).

.....

.....

[1]

Section B

Question 6 (12 Marks)

- (a) (i) GUI is a type of environment provided by Operating Systems.
What does GUI stand for?

..... [1]

- (ii) Give **two** features of a GUI.

1.

2.[2]

- (iii) How is a GUI useful to users?

.....

..... [1]

- (iv) Give **two** tasks that are carried out by an Operating System.

1.

.....

2.

..... [2]

(b) Mail merge is a facility provided by a word processing program.

(i) What is **mail merge**?

.....

.....

.....

.....

[2]

(ii) **Circle** the tab which must be clicked to start mail merge.

File Home Insert Design Layout References Mailings Review View Help

[1]

(iii) What are the **three** steps of a mail merge process?

1.

.....

2.

.....

3.

.....

[3]

Question 7 (14 Marks)

XYZ Hardware Company has two offices, one in Port Louis and one in Dubai. The company has decided to use a VPN to connect the two offices.

- (a) (i) What does the acronym **VPN** stand for?

..... [1]

- (ii) Give **one** reason why the company is using a VPN.

.....
..... [1]

- (b) Give **three** network components for the proposed VPN.

1.
2.
3. [3]

(c) The company decides to use a star topology for its branch in Port Louis.

(i) Draw a **labelled diagram** of a star topology.

[3]

(ii) Give **one** disadvantage of using a star topology.

.....

.....

[1]

- (d) The company decides to create a website.
Use the HTML codes below to show what will be displayed on the Homepage below.

```
<html>
<head>
<title>HOMEPAGE</title>
</head>

<body>

<p>XYZ HARDWARE COMPANY</p>
<p> We are at your service</p>

</body>
</html>
```



[3]

- (e) Employees of the company should avoid plagiarism.

State **two** ways to avoid plagiarism.

1.
.....

2.
.....

[2]

Question 8 (10 Marks)

A spare parts shop uses Microsoft Excel to record the following information.

	A	B	C	D	E	F
1	XYZ SPARE PARTS SHOP					
2	Part Number	Quantity in Stock	Buying Price (Rs)	Selling Price (Rs)	Profit (Rs)	Availability
3	3875489C	6	7500	9850	2350	Yes
4	1256376S	0	1200	2500		No
5	4632562F	9	5200	7500		Yes
6	1567595A	0	1800	2400		No
7	2359187B	5	2250	3000		Yes
8						

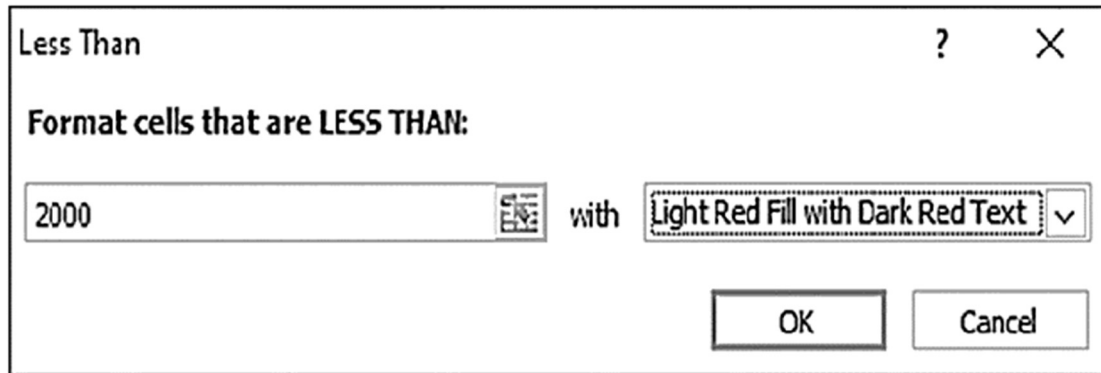
- (a) Write down **one** cell address that has **Wrap Text** applied to it.

..... [1]

- (b) Write down the formula that has been used to calculate the profit in cell **E3**.

..... [2]

- (c) A conditional formatting rule is applied to cell range C3:C7 below.



Write down the **two cell addresses** that will be formatted.

..... [2]

- (d) Write down **a formula** to display the number of different types of spare parts (Part Number) recorded for sales in the shop.

..... [2]

- (e) Write down **a formula** to display the number of spare parts that are not available in stock.

..... [3]

Question 9 (11 Marks)

(a) A flowchart to add a new member to a library database works as follows:

- Input member's ID
- Check if ID exists
- If ID exists, display "Member already exists!"
- End the 'add new member' process
- Else input member details
- Save member record in the library database
- End the 'add new member' process

Draw the flowchart for the above scenario.

[6]

(b) What do you understand by the term **dry run**?

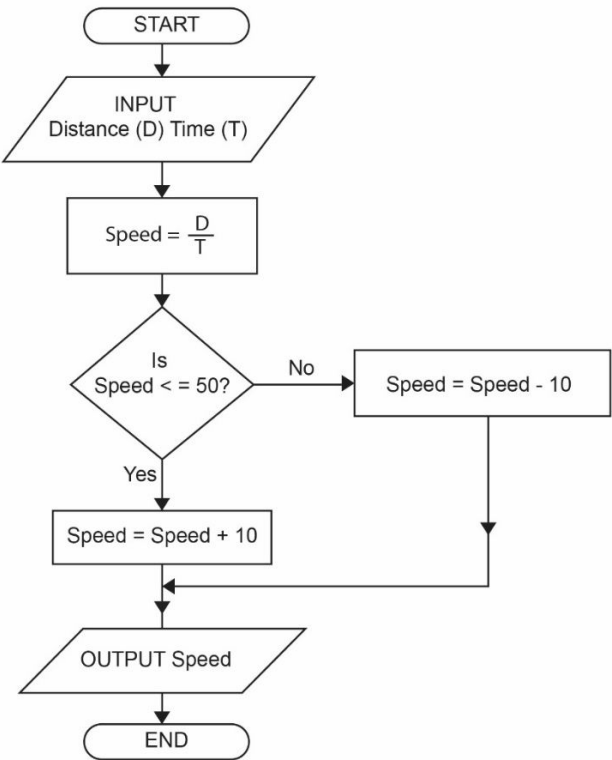
.....

.....

.....

[2]

(c) Study the flowchart below carefully.



State the output for each of the values input for D and T.

INPUT		OUTPUT
D	T	Speed
480	10	
1000	20	
4800	60	

[3]

- END OF PAPER -